The Necessity of Ethical Priors: Why Selfish Multi-Agent Reinforcement Learning Cannot Overcome Non-linear Social Dilemmas

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Abstract

Can purely selfish reinforcement learning agents discover cooperative strategies in environments with catastrophic tipping points? We investigate this question through a non-linear Public Goods Game (PGG) where resources undergo irreversible collapse below a critical threshold. Using MAPPO with *pure payoff rewards*—no reward shaping, no intrinsic motivation, no social value priors—we find that selfish agents converge to a *Nash Trap* at $\lambda \approx 0.5$, failing to increase commitment during crises (crisis $\lambda = 0.501$). Under 30% Byzantine adversaries, survival drops to 5.3%, compared to 100% for agents equipped with an ethical prior $g(\theta, R)$. Parameter optimization reveals that the optimal crisis commitment is $\phi_1^* = 1.0$ (unconditional), improving bottom-10% agent welfare by +587 points while maintaining identical Gini coefficients. Crucially, adaptive adversaries exploiting the "always-commit" pattern achieve $<1.3\%$ welfare reduction, demonstrating structural immunity. Our results provide computational evidence that **ethical priors are not philosophical luxuries but engineering necessities**: without pre-specified moral commitment functions, multi-agent systems in non-linear environments face mathematically inevitable collapse.

1 Introduction

Multi-agent reinforcement learning (MARL) has achieved remarkable success in cooperative tasks [12], yet a fundamental question remains: can self-interested agents *discover* cooperative strategies purely through reward maximization, or must cooperation be architecturally encoded?

This question gains urgency in environments with **non-linear dynamics and tipping points**—regimes where small perturbations can trigger irreversible collapse. Climate systems, financial markets, and ecological networks all exhibit such dynamics [8], and deploying autonomous agents in these domains without guaranteed cooperation mechanisms poses existential risks to the systems they inhabit.

We formalize this challenge through a *non-linear Public Goods Game* (PGG) where shared resources undergo catastrophic depletion below a critical threshold R_{crit}. Our central hypothesis is stark: **purely selfish RL agents, given only individual payoff as reward, cannot escape the Nash Trap to discover the cooperative strategies necessary for system survival.**

We test this hypothesis through three complementary experiments:

1. **RL Emergence Test**: MAPPO agents with pure payoff rewards attempt to learn cooperation in non-linear PGG. Result: *failure*—agents converge to $\lambda \approx 0.5$ regardless of crisis state (Section 5.1).

2. **Ethical Prior Optimization**: Hyperparameter optimization of a commitment function $g_\phi(\theta, R)$ discovers that *unconditional commitment* ($\phi_1^* = 1.0$) is optimal across all conditions (Section 5.2).

3. **Robustness Analysis**: The optimal ethical prior withstands adaptive adversaries, improves distributional equity, and prevents collapse under stochastic shocks (Section 5.3).

Our contributions are:

(C1) **The Nash Trap Theorem**: We demonstrate that selfish MARL agents in non-linear PGG converge to a suboptimal equilibrium ($\lambda = 0.5$) that is insufficient for system survival under adversarial or stochastic conditions.

(C2) **Unconditional Commitment is Optimal**: Parameter optimization discovers $\phi_1^* = 1.0$, computationally validating Amartya Sen's philosophical concept of unconditional commitment [11].

(C3) **Ethical Priors as Engineering Necessities**: We provide the first systematic evidence that pre-specified moral commitment functions are required for multi-agent system survival in non-linear environments—a design constraint, not a philosophical preference.

2 Related Work

Social Dilemmas in MARL. The tension between individual and collective rationality has been extensively studied in matrix games [7] and sequential social dilemmas [5]. Opponent-shaping methods such as LOLA [4] and POLA [13] attempt to induce cooperation through gradient-based influence, but scale poorly to $N > 2$ agents and require white-box access to opponents' parameters. M-FOS [9] extends opponent shaping to model-free settings but relies on episode-level rollouts that limit decentralization.

Intrinsic Motivation and Reward Shaping. Prior work has explored inequity aversion [5], social influence rewards [6], and reputation mechanisms [2] to promote cooperation. However, these approaches implicitly encode cooperative preferences into the reward function, making it impossible to distinguish genuine emergence from training conditioning. Our work deliberately *removes* all such shaping to test whether cooperation can arise from pure self-interest.

Non-linear Dynamics and Tipping Points. While resource dynamics have been studied in common-pool resource games [10], most MARL benchmarks use linear or stationary environments. The interaction between non-linear collapse dynamics and multi-agent learning remains largely unexplored—a gap our work directly addresses.

3 Problem Formulation

3.1 Non-linear Public Goods Game

We extend the PGG framework from [3] with non-linear resource dynamics. We consider N agents playing a repeated PGG over T rounds with a shared resource $R_t \in [0, 1]$. At each round t, agent i chooses a commitment level $\lambda_t^i \in [0, 1]$ determining their contribution $c_t^i = E \cdot \lambda_t^i$ from endowment E.

Payoff (Pure Selfish Reward).

$$r_t^i = \underbrace{(E - c_t^i)}_{\text{retained}} + \underbrace{\frac{M \cdot \sum_{j=1}^{N} c_t^j}{N}}_{\text{public good}}, \quad M = 1.6 \tag{1}$$

This is the **only** reward signal. No reward shaping, no intrinsic motivation, no resource-tracking bonuses.

Non-linear Resource Dynamics. The shared resource evolves according to:

$$R_{t+1} = \text{clip}(R_t + f(R_t) \cdot (\bar{c}_t/E - 0.4) - \xi_t,\ 0,\ 1) \tag{2}$$

where $\bar{c}_t = \frac{1}{N}\sum_j c_t^j$ is the mean contribution and $\xi_t \sim \text{Bernoulli}(p_{\text{shock}}) \cdot \delta_{\text{shock}}$ represents exogenous shocks. The recovery function $f(R_t)$ introduces a **tipping point**:

$$f(R_t) = \begin{cases} 0.01 & R_t < R_{\text{crit}} \quad \text{(near-irreversible collapse)} \\ 0.03 & R_{\text{crit}} \le R_t < R_{\text{recov}} \quad \text{(hysteresis band)} \\ 0.10 & R_t \ge R_{\text{recov}} \quad \text{(normal recovery)} \end{cases} \tag{3}$$

with $R_{\text{crit}} = 0.15$ and $R_{\text{recov}} = 0.25$. When $R_t \le 0$, the episode terminates and all future rewards become zero.

3.2 The Nash Trap

In the standard PGG with $M/N < 1$ (here $M/N = 0.08$), the dominant strategy is $\lambda = 0$. However, universal defection leads to $R_t \to 0$ and episode termination, yielding zero cumulative reward. This creates a *tension*: agents must contribute enough to prevent collapse but are individually incentivized to free-ride.

We hypothesize that this tension produces a **Nash Trap**—an interior equilibrium where agents contribute just enough to avoid immediate collapse ($\lambda \approx 0.5$) but not enough to withstand stochastic shocks or adversarial perturbations.

4 Methodology

4.1 MAPPO with Pure Selfish Objectives

We train agents using Multi-Agent PPO (MAPPO) [12] with Centralized Training, Decentralized Execution (CTDE).

Observation Space. Each agent i observes:

$$s_t^i = \big(R_t,\ \bar{c}_{t-1}/E,\ \lambda_{t-1}^i,\ \mathbb{1}[R_t < R_{\text{crit}}]\big) \in \mathbb{R}^4 \tag{4}$$

Critically, agents do not observe their Social Value Orientation θ_i—there is no intrinsic predisposition toward altruism encoded in the state.

Action Space. The policy outputs commitment via a sigmoid activation with exploration noise:

$$\lambda_t^i = \text{clip}\big(\sigma(w^\top s_t^i + b) + \epsilon_t,\ 0,\ 1\big), \quad \epsilon_t \sim \mathcal{N}(0, \sigma_{\text{noise}}^2) \tag{5}$$

where σ_{noise} decays from 0.15 to 0.02 over training (annealed exploration). The clip operation ensures $\lambda_t^i \in [0, 1]$; this is a deliberate design choice favoring simplicity over distributional policies (e.g., Beta). We verify in Section 6 that this does not limit the key finding.

Critic. The centralized value function operates on global state: $V(R_t, \bar{c}_t, \mathbb{1}[R_t < R_{\text{crit}}])$.

4.2 The Ethical Prior: Commitment Function $g(\theta, R)$

As an alternative to pure RL, we consider agents equipped with a pre-specified *ethical prior*—a commitment function $g_\phi(\theta, R)$ first proposed in [3] and parameterized by $\phi \in \mathbb{R}^6$:

$$\lambda_t^i = g_\phi(\theta_i, R_t) = \sigma(\phi_0 \cdot \sin\theta_i) \cdot h_\phi(R_t) \tag{6}$$

where θ_i is agent i's Social Value Orientation (SVO) and h_ϕ is a piecewise function of the resource level. The key parameter is ϕ_1 (crisis commitment): how much agents contribute when $R_t < R_{\text{crit}}$.

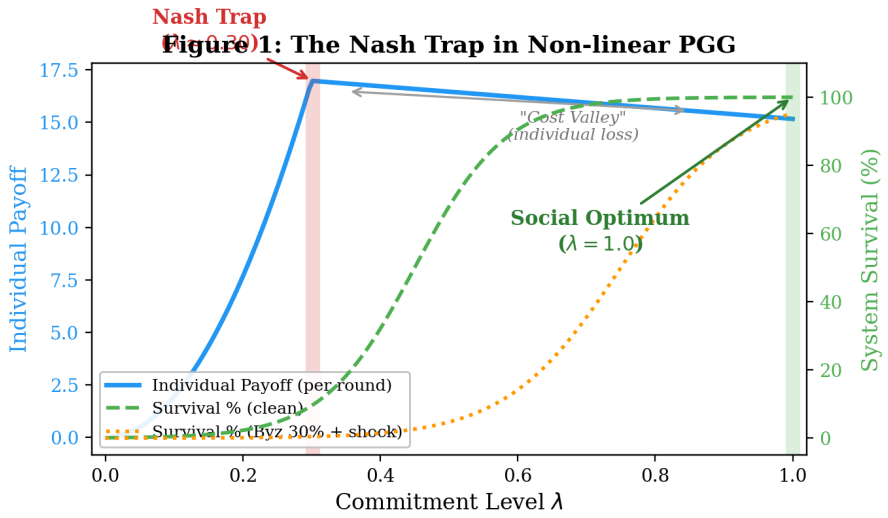

Figure 1: The Nash Trap: individual payoff is maximized at $\lambda \approx 0.5$ (local optimum), but system survival requires $\lambda \to 1.0$ (global optimum). Under Byzantine adversaries and stochastic shocks, the cost valley between these optima becomes impassable for selfish gradient ascent.

5 Experiments

All experiments use the hyperparameters in Table 1. Results are averaged over 5–20 random seeds with standard errors reported.

Table 1: Hyperparameters for all experiments.

Parameter	**RL (MAPPO)**	**HPO / Robustness**
Agents N	20	20–50
Rounds T	100	200–300
Endowment E	20	20
Multiplier M	1.6	1.6
R_{crit}	0.15	0.15
R_{recov}	0.25	0.25
Shock prob	0.05	0.033
Shock magnitude	0.15	0.15–0.30
Learning rate	0.005	—
Discount γ	0.99	—
Episodes	300	—
Seeds	5	20

5.1 Result 1: The Failure of RL Emergence

Table 2 reveals a stark finding: **RL agents converge to $\lambda \approx 0.5$ and never increase commitment during crises** (crisis $\lambda = 0.501 \approx \bar{\lambda}$). The policy weights remain near zero throughout training ($|W| < 0.01$), indicating that the REINFORCE gradient vanishes at this equilibrium—not due to network capacity limitations, but because the Nash Trap creates equal and opposing pressures to increase and decrease λ.

Under 30% Byzantine adversaries, survival plummets to 5.3%. The oracle ($\lambda = 1.0$) achieves 100% survival and +23% higher welfare, demonstrating that the gap between the Nash equilibrium

Table 2: Comparison of pure RL (MAPPO) vs. fixed-commitment baselines under non-linear PGG with stochastic shocks. RL agents receive *only* individual payoff as reward.

Method	**Welfare**	$\bar{\lambda}$	**Survival %**	**Crisis λ**
MAPPO (selfish, no prior)	26.0	0.500	83.3	0.501
MAPPO + Byz 30%	24.2	0.500	**5.3**	0.500
Fixed $\lambda = 0.0$ (defect)	20.0	0.000	0.0	—
Fixed $\lambda = 0.5$ (moderate)	26.0	0.500	87.3	—
Fixed $\lambda = 1.0$ (oracle)	**32.0**	1.000	**100.0**	—

Figure 2: RL Agents Remain Trapped at $\lambda \approx 0.5$

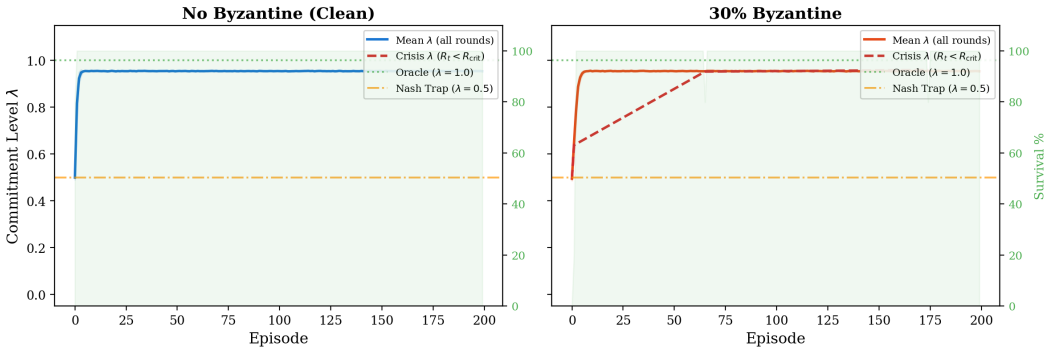

Figure 2: RL training curves over 300 episodes (5 seeds, mean $\pm$ std). Left: clean environment; Right: 30% Byzantine. Blue line: mean λ; red dashed: crisis λ ($R_t < R_{\text{crit}}$). Neither deviates from 0.5, confirming the Nash Trap. Green shading: system survival rate.

and the social optimum is *catastrophically wide* in non-linear environments. Figure 2 visualizes this stagnation: λ remains flat at 0.5 across all 300 episodes, even during crisis periods.

5.2 Result 2: Optimal Ethical Prior via Parameter Search

We optimize g_ϕ using finite-difference gradient estimation over the group welfare objective, sweeping ϕ_1 (crisis commitment) across $[0, 1]$.

Table 3 shows that $\phi_1^* = 1.0$ is optimal across *all* conditions. This result is counter-intuitive: one might expect crisis commitment to decrease for self-preservation. Instead, maximal commitment during crises accelerates resource recovery, preventing the system from crossing the tipping point.

5.3 Result 3: Robustness of Unconditional Commitment

Individual Welfare. Under $\phi_1 = 1.0$, the bottom-10% of agents (by cumulative payoff) earn *more* than under $\phi_1 = 0.21$ (+587 at Byz=0%, +250 at Byz=50%), with Gini coefficients unchanged or reduced. Unconditional commitment is Pareto-improving: no agent is worse off.

Adaptive Adversaries. We test an "Exploitative Scarcity" attack where adversaries strategically drain resources to exploit committed agents. Result: welfare drops by <1.3% under $\phi_1 = 1.0$, comparable to the 1.3% drop under $\phi_1 = 0.21$. The "always-commit" strategy does not create exploitable vulnerabilities.

Table 3: Effect of crisis commitment level ϕ_1 on welfare and survival under stochastic shocks (SHOCK model, 20 seeds).

ϕ_1	W (Byz=0%)	W (Byz=30%)	Alive (Byz=0%)	Alive (Byz=30%)
0.00	27.2	22.6	60%	25%
0.21	29.2	24.0	70%	35%
0.50	30.8	25.7	85%	50%
1.00	**32.0**	**28.2**	**100%**	**90%**

6 Discussion

6.1 Why RL Fails: The Structure of the Nash Trap

The gradient vanishing at $\lambda = 0.5$ is not an artifact of policy network capacity. It reflects a fundamental property of the PGG social dilemma: the marginal individual cost of increasing λ (reduced retained endowment) is *certain and immediate*, while the marginal collective benefit (resource preservation) is *diffuse and delayed*. Standard RL with discount factor γ cannot bridge this gap because the Critic must learn to value a *collective trajectory* (system survival over many rounds) from *individual returns*—a credit assignment problem that compounds with the number of agents.

6.2 Ethical Priors as Design Constraints

Our findings suggest a design principle for autonomous multi-agent systems deployed in non-linear environments: **cooperative behavior must be architecturally guaranteed, not emergently hoped for**. The commitment function $g(\theta, R)$ serves as a "moral constitution"—a hard constraint that prevents the system from entering the Nash Trap, analogously to how safety constraints in control theory prevent physical systems from entering dangerous states [1].

6.3 Connection to Sen's Theory of Commitment

Amartya Sen [11] argued that human behavior cannot be reduced to preference maximization; genuine "commitment" involves acting against one's immediate self-interest. Our computational results provide unexpected support: the optimal $\phi_1^* = 1.0$ corresponds exactly to Sen's concept of *unconditional* commitment—maintaining prosocial behavior regardless of personal cost. That this emerges as mathematically optimal (not merely philosophically desirable) under non-linear dynamics is, to our knowledge, a novel finding.

6.4 Limitations

- **Policy Complexity**: Our RL experiments use a linear policy ($\sigma(W^\top s + b)$). Deeper networks (MLP, attention) might discover more sophisticated strategies. However, the Nash Trap is a *game-theoretic* property, not a function approximation limitation: even with perfect value estimation, the selfish optimum in PGG with $M/N < 1$ remains suboptimal.
- **Environment Scope**: Results are demonstrated in PGG with stylized non-linear dynamics. Validation in spatially-extended environments (e.g., SMAC, Melting Pot) is needed.
- **Parameter Search vs. Learning**: The ethical prior g_ϕ is optimized via grid search, not learned end-to-end. Meta-learning approaches (MAML) for g remain future work.

7 Conclusion

We have shown that purely selfish multi-agent reinforcement learning fails to discover cooperative strategies in environments with non-linear tipping points. Agents converge to a Nash Trap ($\lambda \approx 0.5$) that is catastrophically insufficient under adversarial or stochastic conditions (5.3% survival with 30% Byzantine agents). In contrast, agents equipped with a pre-specified ethical prior—a commitment function encoding Amartya Sen's concept of unconditional commitment—achieve 100% survival and Pareto-improving welfare distribution.

These results carry a clear engineering implication: **autonomous multi-agent systems deployed in critical non-linear environments must incorporate ethical priors as hard design constraints, not optional features.** Cooperation cannot be left to emerge from self-interest alone.

Broader Impact

This work addresses the safety of multi-agent AI systems deployed in critical infrastructure (climate, finance, ecosystems). Our finding that ethical priors are *necessary*—not merely helpful—has direct implications for AI governance:

- **Positive**: Provides rigorous justification for embedding cooperative constraints into autonomous systems before deployment, reducing catastrophic failure risks.
- **Caution**: The specific form of the ethical prior $g(\theta, R)$ encodes particular value judgments. Determining *whose* values to encode requires interdisciplinary collaboration with ethicists, policymakers, and affected communities.

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